

claude-windows / 9f6357d8-0e35-491a-a830-052e094c7f37

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\subagents\workflows\wf_635348d6-51c\agent-a7542e1e4d010ad4d.jsonl`  
- SHA-256: `b78c50240d2486f86e6e0959ceefc677332354b7f839ebff46a63cd804ee9e75`  
- Source modified: `2026-06-21T13:55:03+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_635348d6-51c`  
- session_id: `9f6357d8-0e35-491a-a830-052e094c7f37`
```

Transcript

1. user (2026-06-21T13:54:04.284Z)

In the engine repo at C:/Users/avidu/Projects/badminton-highlight-indexer (GitHub Khelsutra/badminton-highlight-indexer), determine whether the Cloud Run / serving track is being ACTIVELY worked on. Use: `git log --oneline -40 origin/master` (recent serving/M-series commits + dates), and ``gh pr list --repo Khelsutra/badminton-highlight-indexer --state open`` and ``--state merged --limit 30`` (filter to serving / Cloud Run / M5?M12 / compute / Dockerfile). List recent MERGED serving PRs and any OPEN ones. Judge: is this actively in progress right now (rapid recent merges, open PRs, a live tracker)? Note the cadence/evidence and the next planned Cloud-Run steps. Return structured JSON.

2. assistant (2026-06-21T13:54:06.973Z)

I'll investigate the Cloud Run / serving track activity. Let me start by pulling the latest and examining the git log and PRs.

3. user (2026-06-21T13:54:10.208Z)

```
e68b583 fix(api): resolve a bucket-ingested video's source to a local file before stitching (#289)  
43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API (default-OFF) (#288)  
8cdf65c feat(serving): M6 ? Cloud Run GPU dispatch shim + worker image (default-OFF, mocked) (#287)  
b5af66e feat(serving): M5 ? truly-local profile per-profile startup checks + L4 profile-parity test (default-OFF) (#286)  
fefaf412 docs(serving): resolve S8 design picks (D16/D17) + reconcile per-video P3 vs M3 (#285)  
06ff4b7 test(api): lock gs://|gcs:// /api/process -> queryable API-DB video_id (B-P1 guard) (#284)  
cee4baa feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays strict) (#283)  
2afd54f feat(serving): M3 ? configurable output/scratch base (de-hardcode output/, default-OFF) (#282)  
41d45fd fix(server): drain async jobs under `python -m backend.main` (double-module hang) (#281)  
e111add feat(serving): M2 ? URI-aware input (input_backend/input_root, local=identity) (#280)  
ce22791 feat(serving): M1 ? deployment.profile + compute_target config (pure, additive, default-OFF) (#279)  
f518e97 docs(serving): M0 APPROVED + lock D13-D15 (switchable profiles, Drive Northstar, consent-not-surprise) (#278)  
49c082a docs(serving): lock the 5 M0 gating decisions (D7-D12) + add the data-flywheel harness (M11) (#277)  
f592281 docs: point I18N_PLAN + UI_PROPOSAL (+ DOC_STATUS) at the khelsutra UX/l10n project (#275)  
f6f15ca docs: isolate field-ops/PMF + Roora into archives/; carve a no-ML docs/data_pipeline/ (#276)  
03fa951 docs: resolve DOC_STATUS findings #1 (per-video supersede) + #2 (NEXT_STEPS refresh)
```

```

(#274)
aa5e9db docs: add DOC_STATUS register + archival due-diligence convention (#273)
bd42250 docs: spin up COMPUTE_DECOUPLED_SERVING subproject + ADR-014 (#272)
4957b72 fix(worker): configure logging + explain a 0-segment infer (the cold-start blind spot)
(#271)
4134721 docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP) (#270)
c8e60ee fix(gen0): provenance-stamp the manifest git_sha off a git-archive tarball (#269)
e6110ce feat(api): GET /api/health + return download_url from /api/compile (M1/P6) (#268)
a555887 docs(deploy): M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed; both VMs deleted ($0) (#267)
b05401b docs: drop DigitalOcean ? GCP is the sole compute stack (ADR-013) (#266)
4d546ec fix(gcp): teardown defaults to DELETE (a stopped VM still bills its disk) (#265)
dd30e9b feat(workspace): per-video workspace convention ? design doc + Phase 0 helper (#264)
dc9111b docs(adr): ADR-012 ? GPU-native compute; TPU deferred; Cloud Run+GPU is the serving
model (#263)
d8c5aa0 docs(deploy): report polish ? highlight reel smoke-test result + VM-deleted final state
(#262)
0d73d96 docs(deploy): first real GCP-L4 run report + gpu_setup conda-ToS/deps fix + Ops Agent
runbook (#261)
380a4d0 docs(decoupled): narrow DECOUPLED_COMPUTE scope (M0?M2 + M3.5); defer M3/M4; DO=train,
GCP=serve (ADR-011) (#260)
36ed29f feat(wasb): native runner defaults to streaming + correct M3.5 parity gate to a
tolerance (#259)
38c2d0f feat(storage): gdrive:// reads a locally MOUNTED Google Drive (no API) (#258)
55ae408 docs(decoupled): GCP pivot runbook + DO?GCP ADR-010 (deploy/gcp + handoff) (#257)
dc53db5 feat(detectors): Linux-native WASB runner (M3.5) + worker real-trajectory extraction
(#256)
5c77f48 docs(decoupled): HANDOFF.md ? session checkpoint / resume-here (#255)
d5407eb feat(worker): Gen-0 train-from-bucket (worker train) on a CPU stub-trajectory mock
(#254)
814d813 feat(deploy): GPU machine provisioning ? NVMe scratch + native WASB setup (#253)
f06c6a6 fix: cross-platform canonicalization (Linux/macOS, not just Windows) (#252)
6e5ac9e feat(worker): python -m backend.worker dispatcher ? bucket-driven train/infer (#251)
d292854 docs: SETUP_NEW_MACHINE ? canonical zero?train/infer runbook (#250)

```

4. user (2026-06-21T13:54:11.027Z)

```

290      feat(serving): M9-auth ? Cloudflare Access edge-auth + per-tenant owner_id (default-OFF)
feat/serving-m9-auth      OPEN      2026-06-21T13:49:06Z

```

5. user (2026-06-21T13:54:11.884Z)

```

289      fix(api): resolve a bucket-ingested video's source to a local file before stitching
fix/compile-resolve-bucket-source      MERGED  2026-06-21T13:29:35Z
288      feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API (default-OFF)
feat/serving-m8-cost-ledger      MERGED  2026-06-21T11:16:57Z
287      feat(serving): M6 ? Cloud Run GPU dispatch shim + worker image (default-OFF, mocked)
feat/serving-m6-cloudrun-dispatch      MERGED  2026-06-21T10:53:04Z
286      feat(serving): M5 ? truly-local profile per-profile startup checks + L4 profile-parity
test (default-OFF)      feat/serving-m5-truly-local-checks      MERGED  2026-06-21T10:27:06Z
285      docs(serving): resolve $8 design picks (D16/D17) + reconcile per-video P3 vs M3
docs/serving-design-picks-p3-reconcile      MERGED  2026-06-21T09:58:45Z
284      test(api): lock gs://|gcs:// /api/process ? queryable API-DB video_id (B-P1 guard)
test/gcs-process-queryable-coverage      MERGED  2026-06-21T09:35:11Z
283      feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays
strict) feat/serving-m4-device-context      MERGED  2026-06-21T09:12:12Z
282      feat(serving): M3 ? configurable output/scratch base (de-hardcode output/, default-OFF)
feat/serving-m3-output-base      MERGED  2026-06-21T08:39:30Z
281      fix(server): drain async jobs under `python -m backend.main` (double-module hang)
fix/engine-double-module-job-hang      MERGED  2026-06-21T08:22:35Z
280      feat(serving): M2 ? URI-aware input (input_backend/input_root, local=identity)
feat/serving-m2-input-backend      MERGED  2026-06-21T08:15:18Z
279      feat(serving): M1 ? deployment.profile + compute_target config (pure, additive, default-
OFF) feat/serving-m1-deployment-profile      MERGED  2026-06-21T07:34:37Z
278      docs(serving): M0 APPROVED + lock D13-D15 (switchable profiles · Drive Northstar ·
consent-not-surprise)      docs/m0-approved-d13-d15      MERGED  2026-06-21T06:59:16Z

```

277 docs(serving): lock the 5 M0 gating decisions (D7-D12) + data-flywheel harness (M11)
docs/lock-m0-decisions MERGED 2026-06-21T06:42:05Z
276 docs: isolate field-ops/PMF + Roora ? archives/; carve a no-ML docs/data_pipeline/
docs/restructure-field-roora-data MERGED 2026-06-21T04:48:51Z
275 docs: point I18N_PLAN + UI_PROPOSAL (+ DOC_STATUS) at the khelsutra UX/l10n project
docs/point-i18n-ui-to-khelsutra MERGED 2026-06-21T04:40:42Z
274 docs: resolve findings #1 (per-video supersede) + #2 (NEXT_STEPS refresh)
docs/findings-1-2-pervideo-nextsteps MERGED 2026-06-21T03:28:01Z
273 docs: DOC_STATUS register + archival due-diligence (honestly update ? then archive)
docs/doc-status-register MERGED 2026-06-21T03:12:26Z
272 docs: COMPUTE_DECOUPLED_SERVING subproject + ADR-014 (one pure core, profile-selected
compute) docs/compute-decoupled-serving MERGED 2026-06-21T02:51:08Z
271 fix(worker): configure logging + explain a 0-segment infer (cold-start blind spot)
fix/infer-observability MERGED 2026-06-21T01:16:17Z
270 docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP) docs/archive-
decoupled-v2 MERGED 2026-06-21T00:49:52Z
269 fix(gen0): provenance-stamp the manifest git_sha off a git-archive tarball
fix/gen0-git-sha-provenance MERGED 2026-06-21T00:48:10Z
268 feat(api): GET /api/health + download_url from /api/compile (M1/P6)
feat/api-m1-health-downloadurl MERGED 2026-06-20T18:10:26Z
267 docs(deploy): M2 finalized ? weights/0.1.0-l4 pushed; both VMs deleted (\$0)
docs/m2-finalized-report MERGED 2026-06-20T18:06:35Z
266 docs: drop DigitalOcean ? GCP is the sole compute stack (ADR-013) docs/drop-
digitalocean-gcp-only MERGED 2026-06-20T17:37:21Z
265 fix(gcp): teardown defaults to DELETE ? a stopped VM still bills its disk fix/gcp-
teardown-default-delete MERGED 2026-06-20T17:22:01Z
264 feat(workspace): per-video workspace convention ? design doc + Phase 0 helper
feat/per-video-workspace MERGED 2026-06-20T17:16:11Z
263 docs(adr): ADR-012 ? GPU-native compute; TPU deferred; Cloud Run+GPU serving model
docs/adr-012-gpu-cloudrun MERGED 2026-06-20T17:05:06Z
262 docs(deploy): report polish ? highlight smoke-test result + VM-deleted final state
docs/deploy-report-polish MERGED 2026-06-20T15:22:08Z
261 docs(deploy): GCP-L4 deploy report + gpu_setup conda-ToS/deps fix + Ops Agent runbook
docs/gcp-deploy-report MERGED 2026-06-20T15:14:31Z
260 docs(decoupled): narrow scope to M0?M2 + M3.5; defer M3/M4; D0=train, GCP=serve
(ADR-011) docs/decouple-scope-defer MERGED 2026-06-20T14:13:11Z

6. assistant (2026-06-21T13:54:14.458Z)

The picture is very clear. Let me check the serving tracker doc to confirm the next-planned steps, since the M-series milestones reference a tracker.

7. user (2026-06-21T13:54:15.695Z)

No files found

8. user (2026-06-21T13:54:15.697Z)

No files found

9. user (2026-06-21T13:54:19.096Z)

Found 12 files

docs\CLOUD_GPU_DRYRUN_GUIDE.md
docs\COMPUTE_DECOUPLED_SERVING\README.md
docs\DOC_STATUS.md
docs\NEXT_STEPS.md
docs\archives\decisions\DECISIONS.md
docs\SETUP_NEW_MACHINE.md
docs\GOLDEN_REGRESSION_FIXTURES.md
docs\README.md
docs\COMPUTE_DECOUPLED_SERVING\runlogs\README.md
docs\COMPUTE_DECOUPLED_SERVING\TESTING_STRATEGY.md
docs\archives\past_projects\DECOUPLED_COMPUTE\HANDOFF.md

10. user (2026-06-21T13:54:19.166Z)

No files found

11. assistant (2026-06-21T13:54:22.169Z)

The tracker is `docs/COMPUTE\_DECOUPLED\_SERVING/README.md`. Let me read its progress tracker section.

12. user (2026-06-21T13:54:23.089Z)

```
1      # Compute-Decoupled Rally Serving ? one pure core, profile-selected compute + storage
2
3      > **Status:** ? **M0 APPROVED** (owner, 2026-06-21) ? **IN PROGRESS (build M1+)** .
**Owner:** Avi . **Created:** 2026-06-21 . **Last updated:** 2026-06-21
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to
`docs/archives/past_projects/` when DONE)
5      > **Tracking anchors:** $7 progress tracker is the source of truth; pointer in memory
`current-state` + `docs/NEXT_STEPS.md` once work starts.
6      > **Relation to existing docs:** the productionization successor to the archived
`DECOUPLED_COMPUTE` (M3/M4); realizes the `ComputeBackend`/`StorageBackend` registries from
[PLATFORM_ARCHITECTURE.md](../PLATFORM_ARCHITECTURE.md) $3; the **truly-local** profile is the
L0 tier of [VIDEO_LOCALITY_MODEL.md](../VIDEO_LOCALITY_MODEL.md) and the `LocalDesktop` backend
of [DESKTOP_APP_PLAN.md](../DESKTOP_APP_PLAN.md).
7      > **Naming:** this was the "Cloud Run + GPU serving subproject" (ADR-012). Renamed
**compute-decoupled serving** because **truly-local is a co-equal, first-class profile**, not an
afterthought.
8      > **Honesty note:** claims tagged `[verified]` (checked against code by the design
workflow `wf_56274ff0`), `[design]` (proposed), `[researched]`.
9
10     ---
11
12     > **? Northstar (D14):** a **mobile, app-like** flow ? point at a **Google Drive** video
? **fully cloud-native** run ? the stitched, **ready-to-share** reel lands **back in Drive, next
to the source**. Two design promises around it: a user can **switch local?cloud anytime** (D13,
a headline feature), and execution is **never a surprise** ? auto-detect *suggests*, the user
*confirms* (D15).
13
14     ## 0. TL;DR
15     The rally engine is a **pure function of `(input bytes + config) ? outputs`** in three
deterministic stages ? **DETECT ? WINDOW ? STITCH** ? that **no deployment profile may branch
on**. A **`ComputeBackend` (deployment profile)** decides *where* the core runs (a local process
vs a Cloud Run GPU job); a **`StorageBackend`** decides *where bytes come from* (local FS / USB
/ mounted Google Drive / bucket). **One startup config** picks both. We ship **truly-local**
(local GPU + in-process stitch, zero cloud) and **cloud-serving** (upload `<2GB` / gdrive /
bucket ? a Cloud Run GPU job runs detect **and** stitch, so stitch's compute is metered), plus
**cpu-mock** for CI. The single most important caveat: detection is **byte-identical only on the
same GPU** (cross-GPU is sub-pixel-different, per the 2026-06-20 deploy report) ? so the
**parity guarantee is enforced at the WINDOW + STITCH layer**, not detector-CSV bytes.
16
17     ## 1. Problem & goal
18     - **Why now:** DECOUPLED_COMPUTE proved compute *portability* (M0?M2 + M3.5 on a GCP L4)
and is archived; its deferred **M3 (serving endpoint + per-video billing) + M4 (scale-to-zero +
cost guard)** become this project. The owner needs **both** a hosted service *and* a truly-local
mode, with the **core (detect+stitch) unaffected** by which one runs.
19     - **The two user-facing outcomes** (owner, 2026-06-21):
20     - ***(a) Hosted service** ? operate on small **local uploads (`<2GB`)** *and* load from
**gdrive / a bucket**, spinning up **Cloud Run for both rally segmentation AND stitching**
(stitching is compute-intensive ? run it where its cost is accounted for).
21     - ***(b) Truly-local** ? on a local machine with videos on **local drives/USB + a local
GPU**, run inference **and** stitching **completely locally**.
22     - **"Good" looks like:** the same `video ? rallies ? reel` core gives identical windows
```

+ stitched ranges whether it runs on a laptop or Cloud Run; switching is a **config/startup choice**, never a code branch in the core.

23

24 **## 2. Decisions locked**

25

26 | # | Decision | Source | Implication |

27 |---|-----|-----|-----|

28 | D1 | The core is a **pure 3-stage function** (DETECT/WINDOW/STITCH); **no profile** branch inside it. | design `[verified]` | All profile difference lives in config + the two registries. |

29 | D2 | **Two registries:** `ComputeBackend` (where) selected by `deployment.profile` + `compute_target`; `StorageBackend` (what-bytes) by `input_backend`/`storage.backend`. | ADR-014 | Realizes `PLATFORM_ARCHITECTURE`'s `local`/`hosted`/`byo_worker`. |

30 | D3 | **Profiles shipped:** `truly-local`, `cloud-serving`, `cpu-mock` (CI); `byo_worker` reserved/deferred. | ADR-014 | Enum/seam reserved so BYO isn't precluded. |

31 | D4 | In **cloud-serving**, STITCH runs on Cloud Run (co-located with detect) so CPU/wall + egress are metered into the per-job cost ledger. | owner (a) | Stitch-on-laptop would hide real cost. |

32 | D5 | **Parity enforced at WINDOW+STITCH** (byte-exact for a fixed trajectory); **detect** is byte-exact only same-GPU. | deploy report 2026-06-20 | Cross-GPU asserted at the rally/window level, never CSV bytes. |

33 | D6 | **Default-OFF**, smallest-safe-first, one PR per milestone; any core-caller change (e.g. configurable output base) ships behind a flag + a Launch Report. | `[[launch-report-convention]]` | Zero-regression discipline. |

34 | D7 | **Cloud Run cold-start: SCALE-TO-ZERO** ? accept the ~1?3 min cold-start (it's an async job: upload?poll?reel, amortized over the multi-minute job). **No warm pool** (that ~\$500/mo idle would break per-video billing). | owner 2026-06-21 | M6/M7. Revisit a warm lane only on a UX complaint. |

35 | D8 | **Pricebook** = a versioned DB table; cost computed in **USD** (matches GCP billing = one source of truth), **displayed in INR** at a configurable FX rate (Bangalore market); re-versioned when GCP prices change. | owner 2026-06-21 | M8. |

36 | D9 | **Edge auth** = Cloudflare Access (reuse the site's existing CF). Lock the Cloud Run **ingress** to the CF proxy + **verify the Cf-Access JWT signature** (so a direct hit to the Cloud Run URL can't spoof the identity header). | owner 2026-06-21 | M9. One identity system. |

37 | D10 | **Free-tier "data-for-reels" trade:** free reels in exchange for **training rights** (uploaded footage + derived trajectories/labels); **paid tier** = no-train (privacy is the paid feature). **Carve-outs:** train ONLY on attested **ADULT** footage (minor footage ? reel yes, training pool **NO** ? DPDP/GDPR need verifiable parental consent); train on **DERIVED** data + **auto-delete raw**; **ToS counsel-reviewed**; live training-on-uploads **gated to M9** (public signup). Truly-local needs no DPA. | owner 2026-06-21 | M9 + D12; ties `[[paper-coauthor-aim]]` / PERSONALIZATION flywheel. |

38 | D11 | **Quality floor:** Gemini handoff ON* for cloud-serving until the owned model clears the #172 ship-gate (e.g. ?0.70 multi-court) ? then flip via a **Launch Report**; the owned model runs in **shadow/eval** meanwhile. (Gemini = serving/inference ONLY, never a training source ? CLAUDE.md guardrail.) | owner 2026-06-21 | M7 + D12. |

39 | D12 | **Data-flywheel harness** is IN SCOPE (owner add, Q5): capture the (consented, adult) uploaded video + the Gemini reel/rally output ? **reliably store** in the per-video workspace/bucket ? **run shadow evals** (owned-vs-Gemini, dual-eval-set) ? **promote good ones** to the golden TRAINING set (human-reviewed labels) so the owned model climbs toward the D11 floor (then Gemini flips off). | owner 2026-06-21 | new **M11**; ties D10 (consent) + `data_pipeline` + the eval/experiment harness. |

40 | D13 | **Profile** is user-switchable ANYTIME ? a headline **FEATURE**. The same user runs **local** today, cloud tomorrow, back to local ? it's just a config/startup choice; the pure core gives **equivalent** output either way. **Advertise it** ("your machine *or* our cloud ? your call, switch anytime"). | owner 2026-06-21 | Honest caveat: equivalent at the **rally/window** level, not bit-identical across GPUs (D5); a single job runs in one profile, switching is **between** jobs. |

41 | D14 | **NORTHSTAR** ? operate toward this: a **mobile, app-like** flow ? point at a **Google Drive** video ? **fully cloud-native** run ? the stitched, **ready-to-share** reel lands **back in Drive**, next to the source. | owner 2026-06-21 | Implies a `gdrive-api` (OAuth) `StorageBackend` (read source + write reel back) ? distinct from the local **mount** backend; **resolves S8 #7**, new **M12**. The mobile UI is the khelsutra track; the engine must support it (async jobs + status + Drive round-trip / signed URL). Net-new scope (OAuth + Drive write); not necessarily v1, but the design must not preclude it (the `StorageBackend` ABC

accommodates it). |

42 | D15 | ****Auto-detect SUGGESTS, the user CONFIRMS ? execution is NEVER a surprise.**** A GPU owner may still choose cloud; ****cloud (= spend) is never entered silently****; the active profile is always explicit + surfaced. | owner 2026-06-21 | Amends §4.3 (auto-detect = a proposed default, not an auto-decision). |

43 | D16 | ****M6/M7 implementation SHAPE + the cheap §8 picks (owner 2026-06-21):**** (a) ****Cloud Run JOBS**** (not Services-with-GPU) for the detect+stitch unit, ****bucket-poll** for completion** (reuses `JobWaiting`/`claim\_due\_job` verbatim, fits async upload?poll?reel + scale-to-zero D7); (b) ****stitch ENCODE = libx264**** (protects the D5 byte-determinism/parity guarantee + the L4 parity test) ? ****NVDEC for DECODE only**** (a speed lever that never touches output bytes); ****NVENC encode is DEFERRED behind a config knob****, revisit only if M7 cost data justifies relaxing parity (§8 Q8); (c) ****input cap = hard-reject `<2GB`**** for v1 (a config cap so raising to 4GB later is one line; matches the shipped M2 chunked-upload threshold ? §8 Q9). | owner 2026-06-21 | M6/M7. Resolves §8 Q2 (co-located, = D4), Q8, Q9, Q10 (dir name confirmed). |

44 | D17 | ****Real cloud verification BUDGET = \$100 / ?10k cap (owner 2026-06-21):**** the owner authorized actual GCP spend for due-diligence + production-grade testing/verification of M5?M7 (real L4 run, the M5 runlog, the M7 DEPLOY_REPORT), ****provided the cap is respected**** ? confirm exact cmd + hourly cost before each spend, prefer the smallest viable run + cheapest-adequate GPU, and ****DELETE every resource after**** ([[gcp-vm-always-delete]]). Removes the M6/M7 "spend" gate within the envelope; the calibrated pricebook (M8) + counsel ToS (M9) + OAuth app (M12) stay owner/counsel-side. | owner 2026-06-21 | Unblocks the real M5/M7 runs (not just the code) within the cap. |

45

46 ## 3. Foundation ? evolution / ADR lineage ? ***trace the idea end-to-end***

47 This project is the latest step in a long decision chain; each ADR contributed a piece and a ****subtlety that carries forward****:

48

49 | ADR / doc | Contributed | Subtlety carried into this project |

50 |---|---|---|

51 | ****ADR-005**** | Permissive licensing (MIT/Apache-2.0/BSD only) | The engine + any served model + the ****container image**** (ffmpeg, fonts, weights) stay commercial-clean. |

52 | ****ADR-008**** | Owned-model topology; ****train==serve** featurizer single-sourcing** | The "core unaffected across profiles" is the same parity discipline, extended from train/serve to ****local/cloud****. |

53 | ****ADR-009**** | KhelSutra Guru; ****local-first delivery**** | The ****truly-local profile**** is the direct continuation ? privacy/offline self-host stays first-class. |

54 | ****ADR-010**** | DECOUPLED: D0?GCP pivot | The cloud-compute origin (GPU + co-located bucket). |

55 | ****ADR-011**** | DECOUPLED scope = M0?M2+M3.5; ****deferred M3 (serving) + M4 (billing/cost-guard)****; overrode the "rent nothing / not a service" posture | ****This project realizes M3/M4.**** Carries §06's owner-gated gates: ****DPA/consent for non-owner uploads, collector `md5` keying, WASB-C3****. |

56 | ****ADR-012**** | GPU-native; TPU deferred; ****Cloud Run+GPU scale-to-zero = serving model****; ****NVDEC**** named as the decode lever | The serving substrate + the 7-step seed scope this expands. |

57 | ****ADR-013**** | Drop D0; ****GCP sole compute stack****; \$200 D0 credits ? non-compute only | The provider for cloud-serving; D0 Spaces remains only a possible S3 ***storage*** target. |

58 | ****? ADR-014 (new)**** | ****Compute-decoupled serving:**** one pure core, profile-selected compute+storage; stitch-where-metered | This doc. ****Refines**** ADR-011/012, does ****not**** supersede them. |

59 | PLATFORM_ARCHITECTURE.md | The `ComputeBackend`/`StorageBackend` registries | The abstraction this realizes in code. |

60 | VIDEO_LOCALITY_MODEL.md / DESKTOP_APP_PLAN.md | L0 fully-local tier / `LocalDesktop` backend | The truly-local profile = these, productionized. |

61 | 07-COMPUTE-ABSTRACTION (archived) | `DeviceContext`/`DeviceProfile` (designed) | WASB hardcodes `device='cuda'` with ****no CPU fallback**** ? a portability gap M4 closes. |

62 | DEPLOY_REPORT_GCP_2026-06-20 (archived) | First real L4 run; ****cross-GPU sub-pixel parity finding**** | Why D5 (parity at window level, not CSV bytes). |

63

64 ## 4. Design / architecture

65 See [`architecture.svg`](architecture.svg) for the picture. ****The core never changes; the box around it does.****

66

67 ### 4.1 The core contract (must stay byte/behaviour-identical) `[verified]`

68 1. ****DETECT (GPU):**** `DetectorRunner.run_predict(video_path, output_dir, video_id) ?

```

CSV[Frame,Visibility,X,Y]` (`backend/pipeline/detectors/base.py:164`). Three interchangeable
impls ? `WasbRunner` (WSL), `NativeWasbRunner` (Linux GPU), `StubDetectorRunner` (CPU mock) ?
emit the **identical CSV schema**. The trajectory CSV is the stable artifact boundary.
69     2. **WINDOW (CPU pure fn)** `trajectory_to_action_windows(points, fps, frame_width, ?)
? windows` (`tracknet_runner.py:281`). A shared `@staticmethod`, zero GPU/IO/randomness ? same
trajectory + config = byte-identical candidate windows. Used by every runner *and* the
eval/regression stack verbatim.
70     3. **STITCH (CPU/ffmpeg)** `VideoCompiler.compile_highlights(raw_video, segments,
out_name) ? reel` (`stitcher/compiler.py:303`). Pure FS+subprocess (merge ? ffmpeg ? watermark
? per-clip libx264 trim ? concat). Deterministic for a given input + ranges + watermark config.
71
72     **Non-goal (the trap to avoid)** *no* code branch inside detect/window/stitch keyed on
profile ? the same discipline the experiment harness already enforces (a disabled cue is byte-
identical to CONTROL).
73
74     ### 4.2 The deployment profiles
75     | Profile | Compute | Storage | Orchestration | When |
76     |---|---|---|---|---|
77     | **truly-local** | `compute_target=local_gpu`; `detector_impl=native` (Linux) / `auto`
(WSL); stitch **in-process** | `local` (or `gdrive` = mounted Drive) | `python -m backend.main`
or `worker infer` CLI; existing async job worker | Self-host / offline / privacy / dev-on-a-GPU-
box ? owner (b) |
78     | **cloud-serving** | `compute_target=cloud_run`; `detector_impl=native` (L4, cuda);
**detect AND stitch on the same Cloud Run job** | `gcs` (or gdrive mount / assembled upload);
per-video workspace; reels via signed URL | Cloud Run control plane enqueues ? Cloud Run GPU job
pulls/runs/pushes; `WAIT_AND_RESUME` reschedules the control-plane job; scale-to-zero | Hosted
SaaS ? owner (a) |
79     | **cpu-mock** | `compute_target=cpu_mock`; `detector_impl=stub` (synth/replay) |
`local`/in-memory fakes | `run_all_tests.py` / pytest | CI, regression, profile-parity, GPU-free
dev |
80     | *byo_worker* (deferred) | tenant GPU via outbound pull agent | tenant's own bucket
(signed URLs) | long-poll pull; same job table scoped by tenant | Privacy-sensitive enterprise ?
**DEFERRED**, enum/seam reserved |
81
82     ### 4.3 Profile selection (the "startup/setup config") `[design]`
83     ONE new top-level `deployment` section on `AppConfig` + preset application + env auto-
detect in `load_config`:
84     - `deployment.profile ? {truly-local | cloud-serving | cpu-mock}` (empty = today's
manual knobs, **fully backward-compatible**).
85     - Selecting a profile applies a **coherent preset bundle** of *existing* knobs that
today drift independently: **INPUT** (new `input_backend`/`input_root`, parallel to the output-
side `StorageConfig`), **COMPUTE** (new `compute_target` enum locking `detector_impl` +
`skip_ai_handoff` + workspace strategy), **STORAGE**
(`storage.backend`/`workspace_root`/`single_folder_per_video`, all already exist).
86     - **Precedence**: built-in defaults ? profile preset ? `config.json` ? env
(`RALLY_DETECTOR_IMPL`, `WASB_*`, `DEPLOYMENT_PROFILE`, ?) ? worker `--set k=v`. Every knob
stays individually overridable.
87     - **Env auto-detect SUGGESTS, the user CONFIRMS (D15 ? no surprise execution): **the env
gives a *proposed* default (`K_SERVICE` ? cloud-serving; `CI=true` ? cpu-mock; else
`torch.cuda.is_available()` ? truly-local), but the user **explicitly confirms or overrides** it
? a GPU owner may still pick cloud, and **cloud (= spend) is NEVER entered silently**. The
active profile is always surfaced (logged/shown). **Per-profile startup checks fail/warn fast**
(cloud-serving without GCS creds/GPU ? loud error before any job).
88     - The worker (`cmd_infer`/`cmd_train`) is **already profile-agnostic** (reads
`config.storage`, accepts `--config`/`--set`) ? no worker rewrite.
89
90     ### 4.4 Compute placement ? and *why stitch runs on Cloud Run* `[design]`
91     DETECT on GPU everywhere; WINDOW CPU-pure everywhere; STITCH is real CPU work. In
**cloud-serving, stitch is co-located in the Cloud Run job** so `gpu_active_seconds` (detect) +
`wall_seconds` (detect+stitch) + `bytes_out` (reel egress) land on **one metered invocation** ?
the per-job cost ledger. Running stitch on a laptop would leave that cost
**invisible/unattributed** ? exactly what the owner wants to avoid. (Stitch stays synchronous in
the GPU job for the common one-reel case; a queued CPU-only stitch fallback only if compiles get
bursty ? the job table + `claim_due_job` already supports both.)
92

```

```

93     ### 4.5 Reuse map ? most of the seams already exist `[verified]`
94     **Exists:** the 3-stage core; `_resolve_runner` (the single compute seam);
`StubDetectorRunner` (`replay_csv` = byte-exact $0 anchor); `StorageBackend` ABC + 4 backends +
`StorageRef.parse_uri`; the worker CLI (fetch/put, `--config`/`--set`); the async job control
plane (`jobs` table, `claim_due_job` CAS); `ExecutionPolicy WAIT_AND_RESUME`/`JobWaiting`
(carries to Cloud Run dispatch verbatim); the chunked-upload router (`<2GB`); `skip_ai_handoff`
(LocalNoAI); storage fakes; golden fixtures + experiment harness. **Partial:** per-video
workspace helper (default-OFF); `DeviceContext` (designed); edge-auth (v4 `user_id` columns
exist). **Net-new:** the `deployment` section; `input_backend`; configurable output base (kill
hardcoded `output/`); Cloud Run dispatch shim + container image; cost ledger + pricebook; edge-
auth header extraction; startup env detection.
95
96     ### 4.6 The data-flywheel harness (D12 / M11) `[design]`
97     Gemini-on (D11) ships good reels now; this harness is **how we earn the right to flip
Gemini off**. On a **consented, adult** cloud-serving job (the D10 trade), the pipeline
additionally:
98     1. **Captures** the source upload + the Gemini reel + the rally output
(intervals/report) into the per-video workspace (`workspaces/<video_id>/`), instead of the no-
consent auto-delete path.
99     2. **Reliably stores** them durably in the bucket (the consented-retention bucket; raw
video kept only for the training pool, not the no-consent path).
100    3. **Runs shadow evals** ? the owned model scores the same video alongside Gemini
(`backend/eval/experiment.compare` + the dual-eval-set, [[eval-dual-set-regression]]) ? tracks
the owned-vs-Gemini gap per job (the live read on "is the floor reachable yet?").
101    4. **Promotes to golden** ? captured videos that pass review become **golden TRAINING
rows** (human-reviewed labels via the annotation flow ? `data_pipeline/`), growing the corpus ?
the owned model climbs to the D11 floor ? flip Gemini off via a Launch Report.
102    This is the consent?capture?eval?goldenize **flywheel** (the cloud realization of the
PERSONALIZATION contribution model): every consented free reel makes the owned model better.
Reuses `data_pipeline/`, `backend/eval/`, and the M9 consent gate; **adults-only**, raw-
retention gated by the D10 consent.
103
104    ## 6. Honest limits ? what this does NOT do / prove
105    - A green CI suite proves **plumbing + determinism (window+stitch) + profile-parity**
for $0 ? it does **NOT** prove field quality, nor that real-GPU detection is good (that's owner-
side, real-video).
106    - **Cross-GPU detection is NOT byte-identical** ? parity is a window-level claim, not a
CSV-byte claim.
107    - The cloud `0.1.0-l4` weights are **n=2 plumbing-grade** ? cloud-serving ships with
**Gemini handoff ON** until the owned model clears a floor (open Q).
108    - This does not resolve **WASB-C3** (sharing a trained model) or the **DPA/consent**
gate ? both remain owner/counsel-gated before public, non-owner serving.
109
110    ## 7. Deliverables & progress tracker ? **source of truth**
111    Legend: ? Todo . ? In progress . ? Done . ? Gated. **One small PR per row**, off
`origin/master`, no stacking. Default-OFF where it touches core callers.
112
113    | ID | Deliverable | Depends | Gated? | Status | PR |
114    |----|-----|-----|-----|-----|----|
115    | M0 | This design doc + project dir + `runlogs/` + ADR-014 | ? | owner sign-off | ? |
**? APPROVED 2026-06-21** |
116    | M1 | `deployment.profile` + `compute_target` config + preset bundles +
precedence/auto-detect; unit tests (no behaviour change, profile='' default) | M0 | safe | ? |
[#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279) |
117    | M2 | `input_backend`/`input_root`; wire main CLI/API input through
`StorageRef.parse_uri` (local=identity); storage-fake tests | M1 | safe | ? |
[#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280) |
118    | M3 | Configurable output/scratch base ? kill hardcoded `output/`
(`trajectory_hybrid:237,313`, `yolo_hybrid:407`, `gemini`); **DEFAULT stays `output/`**; golden
+ stub-E2E byte-identical | M1 | ? default-OFF + Launch Report on flip | ? |
[#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282) |
119    | M4 | `DeviceContext` + WASB **CPU-fallback**; unified healthcheck/device wiring | M1 |
safe (cuda default unchanged) | ? | [#283](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/283) |
120    | M5 | **truly-local profile** end-to-end (preset + startup checks); first committed

```



```

**runlog** (truly-local sample run) | M2,M3,M4 | owner real-GPU | ? |
[#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286) (preset+startup-
checks+L4 parity ?; runlog ? owner GPU) |
121 | M6 | Cloud Run GPU **dispatch shim** (control plane ? job via storage ? re-claim) +
**containerized worker image** (ffmpeg+CUDA+WASB+fonts); mocked-dispatch tests | M5 | ? owner
(GCP spend) | ? | [#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287)
(shim+image+mocked ?; real build/run = M7 ? owner) |
122 | M7 | **cloud-serving profile** e2e on L4: upload`<2GB` + gdrive/bucket ? detect+stitch
on Cloud Run ? reel via signed URL; **DEPLOY_REPORT** (posterity, sample runs + contact sheet) |
M6 | ? owner (spend) | ? | ? |
123 | M8 | Per-job **cost ledger** (v6 migration: `owner_id, gpu_active_seconds,
wall_seconds, bytes_out, cost_usd, cost_breakdown_json`) + **pricebook** + aggregation +
`api/.../cost` | M7 | ? owner (billing) | ? | [#288](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/288) (ledger+pricebook+/cost ? default-OFF; real prices + usage-emission
? owner/follow-up) |
124 | M9 | **Edge auth** (Cf-Access extraction) + per-tenant scoping on `user_id`;
DPA/consent gate wiring | M7 | ? owner (privacy/counsel) | ? | ? |
125 | M10 | `byo_worker` / zero-infra-BYO ? **design-only, DEFERRED** | M8,M9 | ? explicit
owner approval | ? | ? |
126 | **M11** | **Data-flywheel harness (D12):** on a consented adult job, **capture** the
upload + Gemini reel/rally output ? **reliably store** under the per-video workspace ? **shadow-
eval** owned-vs-Gemini (dual-eval-set / `experiment.compare`) ? **promote** good ones to the
golden TRAINING set (human-reviewed labels via the annotation flow). Closes the loop so the
owned model climbs to the D11 floor. Reuses `data_pipeline/` + `backend/eval/` + the consent
gate (M9). | M7,M9 | ? owner (consent/privacy) | ? | ? |
127 | **M12** | **`gdrive-api` (OAuth) StorageBackend (D14 Northstar):** cloud reads the
source from + writes the stitched reel **back to the user's Google Drive next to the source**
(per-user OAuth consent) ? the mobile/cloud-native Drive round-trip. Distinct from the local
mount backend; slots into the `StorageBackend` ABC. | M7,M9 | ? owner (OAuth/consent) | ? | ? |
128
129 **Testing strategy is a first-class deliverable** ?
[`TESTING_STRATEGY.md`](TESTING_STRATEGY.md). **Run logs / sample runs for posterity** ?
[`runlogs/`](runlogs/).
130
131 ## 8. Open questions ? owner *(blocking-gates vs nice-to-knows)*
132 **? The 5 gating questions are LOCKED (owner, 2026-06-21) ? see §2 D7?D12:**
133 1. ? **Cold-start ? D7** (scale-to-zero, accept cold-start).
134 3. ? **Pricebook ? D8** (versioned DB table; USD internal / INR display).
135 4. ? **Edge auth ? D9** (Cloudflare Access; verify Cf-Access JWT, lock ingress to
proxy).
136 5. ? **DPA/consent ? D10** (free-tier data-for-reels trade; adults-only; derived-data;
counsel-gated to M9).
137 6. ? **Quality floor ? D11** + the **data-flywheel harness D12 / M11** (Gemini-ON until
the owned model clears the floor; capture?store?eval?goldenize meanwhile).
138
139 **? The remaining smaller picks are LOCKED (owner, 2026-06-21) ? see §2 D16:**
140 2. ? **Stitch placement ? co-located** in the detect job (= D4; one metered unit; the
job table supports a future CPU-split non-breaking, revisit only on bursty-compile metrics).
141 7. ? **RESOLVED ? D14:** cloud needs a **`gdrive-api` (OAuth) StorageBackend** (read the
source from + write the reel back to the user's Drive) ? the Northstar's Drive round-trip ?
distinct from the local **mount** backend (which stays for truly-local). New milestone **M12**.
142 8. ? **Codec ? D16:** **libx264 ENCODE** (protect parity) + **NVDEC DECODE-only**
(speed, no output-byte change); NVENC-encode deferred behind a config knob.
143 9. ? **Input cap ? D16:** **hard-reject`<2GB`** for v1, as a config cap (raise to 4GB
later in one line).
144 10. ? **Directory name ? `COMPUTE_DECOUPLED_SERVING/` CONFIRMED final** (already the
name everywhere + in ADR-014); `CLOUD_RUN_GPU_SERVING/` dropped.
145
146 ## 9. Definition of done
147 - [ ] M1?M4 land (config + input + output-base + device layer), each green on golden
cross-OS + experiment no-regression.
148 - [ ] **Profile-parity test green** ? truly-local vs cloud-mock produce byte-identical
windows + segment ranges.
149 - [ ] **truly-local** (M5) and **cloud-serving** (M7) each have a committed **runlog /
DEPLOY_REPORT** with a real sample run.

```

```

150     - [ ] Cost ledger (M8) makes GPU + **stitch** + egress cost visible per job; edge auth
(M9) gates multi-tenant.
151     - [ ] **Data-flywheel harness (M11)** captures?stores?shadow-evals?goldenizes consented
adult jobs; the owned model is on a measured path to the D11 floor.
152     - [ ] ? flip Status DONE, archive to `docs/archives/past_projects/`.
153
154     ## 10. References
155     **ADR lineage:** ADR-005/008/009/010/011/012/013 ? **ADR-014**
(`docs/archives/decisions/DECISIONS.md`). **Strategy:** `PLATFORM_ARCHITECTURE.md`,
`HOSTING_PLAN.md`, `VIDEO_LOCALITY_MODEL.md`, `DESKTOP_APP_PLAN.md`,
`LOCAL_APPLIANCE_ARCHITECTURE.md` (Khelsutra). **Archived predecessor:**
`docs/archives/past_projects/DECOUPLED_COMPUTE/` (esp. `05-COST-ATTRIBUTION.md`, `07-COMPUTE-
ABSTRACTION.md`, `06-RISKS-AND-OPEN-QUESTIONS.md`, `DEPLOY_REPORT_GCP_2026-06-20.md`). **Code
anchors `[verified]`:** `backend/pipeline/detectors/base.py:164`, `tracknet_runner.py:281`,
`stitcher/compiler.py:303`, `segmenters/trajectory_hybrid.py:65,237`, `backend/storage/*`,
`backend/worker/__main__.py:88`, `backend/api/{uploads,job_worker}.py`,
`backend/infrastructure/database.py`. **Design provenance:** workflow `wf_56274ff0` (4 code-
mappers ? synthesis; 58 findings).
156
157     ---
158     ### Changelog
159     - 2026-06-21 ? created from the design workflow output; M0 in progress.
160     - 2026-06-21 ? **M0 APPROVED** ; **M1 landed**
([#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279)):
`deployment.profile` + `compute_target` + preset bundles + precedence/auto-detect-as-suggestion
(D15) on `AppConfig`. Pure/additive, `profile=''` default == today's behaviour (zero core
change); presets touch only existing knobs (no M2 `input_backend` / M3 output-base over-reach).
Suite 1068 green, cov 84.11%.
161     - 2026-06-21 ? **M2 landed** ([#280](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/280)): `input_backend`/`input_root` + URI-aware main API/CLI input via
`resolve_input_uri`/`acquire_local_input` (local = identity passthrough, byte-for-byte
unchanged; bucket/Drive fetched to scratch). Unsupported scheme ? clean 400; `local://` accepted
on the resolved key; hosted-mode rejects non-local URIs; source URI kept for DB/report
provenance. Suite 1105 green, cov 84.84%.
162     - 2026-06-21 ? **M3 landed** ([#282](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/282)): configurable output/scratch base via
`backend/utils/paths.output_base(config)` ? the 3 segmenters' chunk/handoff/temp dirs root under
`output.output_dir` (default `"output"`, byte-identical; default-OFF ? no Launch Report). Fixes
the latent bug where scratch ignored `--output-dir`. Suite 1114 green, cov ~84.9%.
163     - 2026-06-21 ? **M4 landed** ([#283](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/283)): `DeviceContext` + native WASB **CPU-fallback** via a new `device="auto"`
(cuda-if-available-else-cpu); explicit `"cuda"` stays strict (no silent slow-CPU downgrade),
default unchanged. `"auto"` resolves before argv (never reaches `wasb_infer.py`); CUDA probe
cached. Device-selection tested offline; real CPU-inference run owner/GPU-verified. Suite 1126
green, cov 84.96%. **? M1?M4 (the config/input/output/device layer) COMPLETE.**
164     - 2026-06-21 ? **M5+ batch authorized + design picks locked** (owner): the remaining $8
picks resolved ? **D16** (M6 = Cloud Run **Jobs** + bucket-poll; **libx264 encode + NVDEC
decode-only** ; **hard-reject <2GB** input cap; dir name confirmed) and **D17** (real cloud
verification budget = **$100/?10k cap** , delete-after). Owner gave a blanket go to write+merge
the **default-OFF code** for every remaining gated milestone (M5,M6,M8,M9-auth,M9-consent-
cols,M11-plumbing,M12), one isolated PR each, reserving the gate for the real
run/spend/calibration/counsel. Grounded by gate-map workflow `wf_4849400f`.
165     - 2026-06-21 ? **M8 code landed** ([#288](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/288)): per-job **cost ledger** ? `backend/billing.py` (pure `usage ×
pricebook` math, USD internal / INR display) + a **v6 migration** (NULLABLE cost columns on
`jobs` + a versioned `pricebook` table seeded `placeholder-v0` at $0; additive-NONE, idempotent)
+ `record_job_cost`/`get_job_cost`/`get_video_cost` + `GET /api/jobs/{id}/cost` +
`/api/videos/{id}/cost` + `BillingConfig` (**default-OFF**) + a gated `_maybe_record_job_cost`
hook. Adversarial review (`wf_498b24af`) folded a **major silent-GPU-under-bill** (an unpriced
`gpu_type` is now surfaced, not invisible $0) + the **honest no-op** caveat (usage emission is a
follow-up ? cost 0 until wired) + breakdown reconciliation. Suite green, ruff+mypy clean. **? M8
? ? owner inserts real GCP prices (a new pricebook version) + the worker usage-emission is a
follow-up; placeholder ships $0 = no user-facing change. v6 is the shared bump (M9-consent cols
extend it).**
166     - 2026-06-21 ? **M6 code landed** ([#287](https://github.com/Khelsutra/badminton-

```

highlight-indexer/pull/287)): `backend/compute/` ComputeBackend registry ? `CloudRunCompute` dispatches a Cloud Run ****Job**** (injectable client) then ****bucket-polls**** a done-marker via `execution\_policy.poll\_until` + the storage layer (D16); `get\_compute\_backend` returns a backend ONLY for `compute\_target=cloud\_run` (else `None` ? the worker's in-process path, ****default-OFF****). The dispatched container runs ****inline**** (forced `local\_gpu`+native, never re-dispatch); the worker writes a `--done-uri` completion marker on success.

`deploy/cloudrun/{Dockerfile,README}` = the L4 image (CUDA+ffmpeg+fonts+`wasb` conda env; weights staged per WASB-C3) + the M7 runbook. Adversarial review (`wf\_87b22fa4`) folded a ****blocker**** (cloud `fetch` needs a dst ? reads silently returned `{}` on gcs/s3) + a poll-wait test gap + `PolicyTimeout`?clean rc 4. Suite green, ruff+mypy clean. ****? M6 ? ? the real build/push/L4 run + DEPLOY_REPORT is M7 (owner, D17 budget).****

167 - 2026-06-21 ? ****M5 code landed**** ([#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286)): `backend/config/startup.py` ? per-profile fail/warn-fast startup checks (cloud-serving + non-cloud storage = the one fatal coherence error; GPU/creds mismatches = warnings, the runner healthcheck stays the authority), wired into the server (`\_enforce\_startup\_or\_exit` ? exit 1) + the worker (`cmd\_infer`/`cmd\_train` ? rc 2), default-OFF (probe gated behind an active profile ? the worker stays torch-free at `profile=""`). ****L4 profile-parity test**** (`tests/test\_profile\_parity.py`): same stub trajectory ? byte-identical windows + segment ranges across truly-local / cpu-mock / no-profile + worker-vs-in-process. Adversarial 4-lens review folded (`wf\_34671ac1`). Suite green, ruff+mypy clean. ****? M5 ? ? the real-GPU truly-local runlog (L5) is owner-side**** (template pre-staged in `runlogs/`).

168

13. assistant (2026-06-21T13:54:26.032Z)

I have everything I need. Let me check the open PR #290 detail and confirm there's exactly one open serving PR.

14. user (2026-06-21T13:54:27.830Z)

```
{"additions":406,"createdAt":"2026-06-21T13:49:06Z","deletions":13,"isDraft":false,"number":290,"state":"OPEN","title":"feat(serving): M9-auth ? Cloudflare Access edge-auth + per-tenant owner_id (default-OFF)"}
```

15. assistant (2026-06-21T13:54:37.988Z)

I now have a complete and confirmed picture. The Cloud Run / serving track is being very actively worked on right now.

Summary of evidence:

- ****Tracker:**** `docs/COMPUTE\_DECOUPLED\_SERVING/README.md` §7 ? a live progress tracker, status "IN PROGRESS (build M1+)", last updated 2026-06-21, with M0-M12 milestones.
- ****Cadence:**** M0 approved through M8 code all merged on a single day (2026-06-21), plus M9-auth open the same day. Roughly a dozen serving PRs in ~12 hours (#278 at 06:59Z ? #289 at 13:29Z), interleaved with the M2-M8 deploy/decoupled run earlier (#250-#268).
- ****One open serving PR:**** #290 (M9-auth, edge-auth, default-OFF, not draft).
- ****Next planned:**** M7 (real Cloud Run L4 deploy: upload<2GB/gdrive/bucket ? detect+stitch on Cloud Run ? reel via signed URL + DEPLOY_REPORT) is the key owner-gated real-run step; M9 consent gate, M11 data-flywheel, M12 gdrive-api OAuth backend follow.

16. user (2026-06-21T13:54:56.571Z)

Structured output provided successfully

17. assistant (2026-06-21T13:55:03.687Z)

The Cloud Run / serving track is being very actively worked on. Full findings returned via StructuredOutput.

Key facts:

- ****Live tracker:**** `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMPUTE\_DECOUPLED\_SERVING\README.md` (status "IN PROGRESS (build M1+)", §7 progress table is the source of truth, last updated 2026-06-21).
- ****Cadence:**** ~13 serving PRs merged in a single ~7-hour burst on 2026-06-21 (#278 M0-approved

? #288 M8), with M1-M8 code all landing default-OFF, one isolated PR per milestone off
`origin/master`.

- ****Open PR:**** #290 (M9-auth, Cloudflare Access edge-auth, default-OFF, not draft) ? opened the same day.
- ****Next planned:**** M7 (the real GCP L4 Cloud Run e2e + DEPLOY_REPORT, owner/spend-gated under the D17 \$100 budget) is the headline next step; M6 already shipped the Cloud Run dispatch shim + `deploy/cloudrun/Dockerfile` worker image (mocked). Then M8 real-prices/usage-emission, M9 consent gate, M11 data-flywheel, M12 gdrive-api OAuth backend. M10 byo_worker stays deferred.